

AidPulse AI — Smart Disaster Response System

Real-Time Location-Aware Evacuation Navigation, Hospital ER Bed Tracking & Gemini AI 7-Day Risk Prediction

Global Real Road Navigation

OSRM 100% street-snapped routing replacing straight lines with turn-by-turn safe bypass guidance.

Gemini 1.5 Flash Risk AI

Satellite hazard forecasting computing precise 7-day disaster risk scores (LOW, MODERATE, HIGH, CRITICAL).

Live ER/ICU Bed Tracker

OpenStreetMap live hospital node sync, oxygen hours, blood bank units, and 112 direct triage calling.

 PROBLEM STATEMENT

The Chaos of Natural Disasters & Emergency Response Failures

During floods, earthquakes, cyclones, and fires, traditional tools fail citizens when seconds count.

Lack of Real-Time Shelter Awareness

Victims lack visibility into nearest open shelters, available bed capacity, clean water, or power backup status.

Hazardous & Blocked Roads

Standard navigation apps guide evacuees into flooded underpasses, collapsed bridges, or standing water hazards.

Medical Triage Bottlenecks

Emergency rooms overflow while victims cannot locate nearby free ER beds, ICU units, or oxygen reserves.

Delayed Emergency Triage Guidance

Panicked citizens lack immediate access to localized first-aid protocols and personalized AI triage.

 PROJECT AIM & OBJECTIVES

Empowering Humanity with Real-Time Life-Saving Guidance

Our mission is to build a location-aware disaster management ecosystem providing immediate navigation and AI triage.

1. Street-Snapped Safe Navigation

Utilize Open Source Routing Machine (OSRM) to snap driving paths 100% onto actual street roads and bypass flood hazards.

2. Real-Time Medical Triage

Provide live ER/ICU bed metrics and 24/7 direct emergency hotline connections for immediate rescue dispatch.

3. Gemini Predictive Analytics

Harness Google Gemini 1.5 Flash AI to deliver 7-day satellite disaster risk modeling for any global region.

💡 THE PROPOSED SOLUTION

A Unified Emergency Ecosystem Powered by Live Telemetry & AI

AidPulse AI integrates real-world GIS data, live weather metrics, seismic feeds, and Gemini 1.5 Flash AI.

OSRM Real Road Engine

Replaces straight-line polylines with 100+ snapped road geometry points following actual streets, highways, and rotaries.

Route Weather Telemetry

Compares origin vs. destination weather telemetry (Temperature, Rain mm/h, Wind speed) to warn against flash floods.

Live USGS Seismic Telemetry

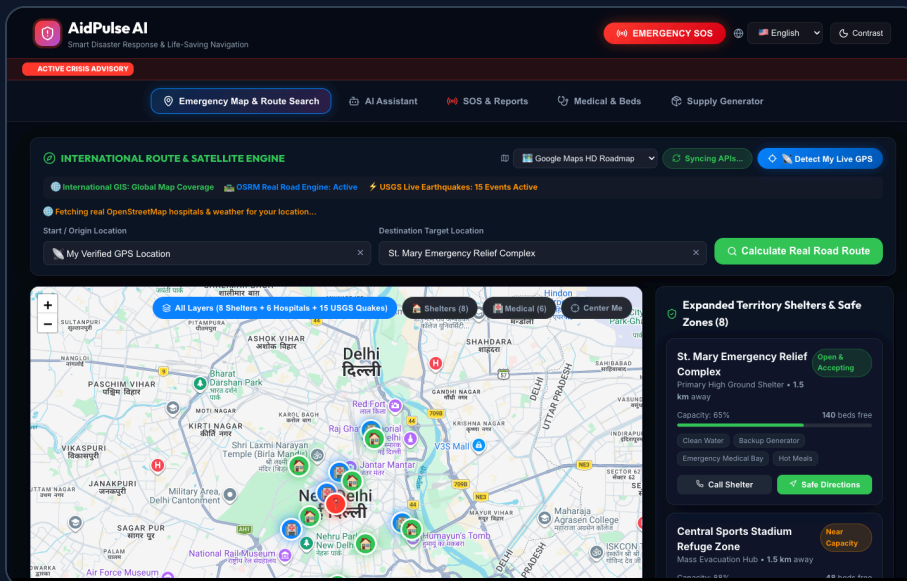
Plots real-time global earthquake feeds with exact Richter magnitude, depth, and official USGS event links.

Multi-Language Accessibility

Fully localized in 6 major global languages: English, Spanish, Hindi, French, Bengali, and Arabic.

FEATURE SHOWCASE 1

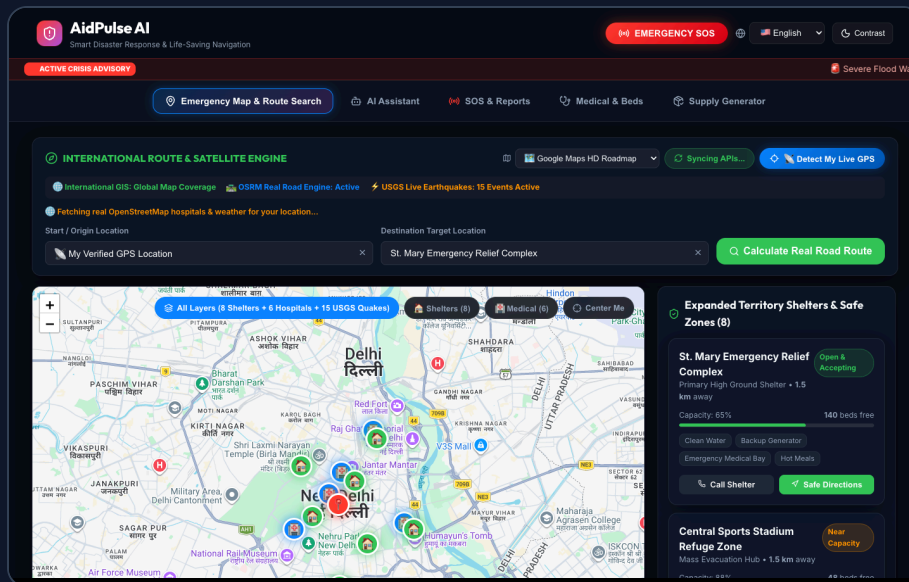
Real-Time Evacuation Map with OSRM Street Navigation



Key Capabilities Demonstrated:

- **OSRM Road Snapping:** Thick green polyline snaps 100% to actual street geometry.
- **Dual Typeahead Search:** Free-text origin and destination search with Nominatim geocoding.
- **Google Maps HD Switcher:** Toggle Google Maps HD Roadmap, Hybrid Satellite, or CartoDB tiles.
- **USGS Seismic Feed:** Live worldwide earthquake markers updated dynamically.

Gemini 1.5 Flash AI Triage & Global Risk Predictor

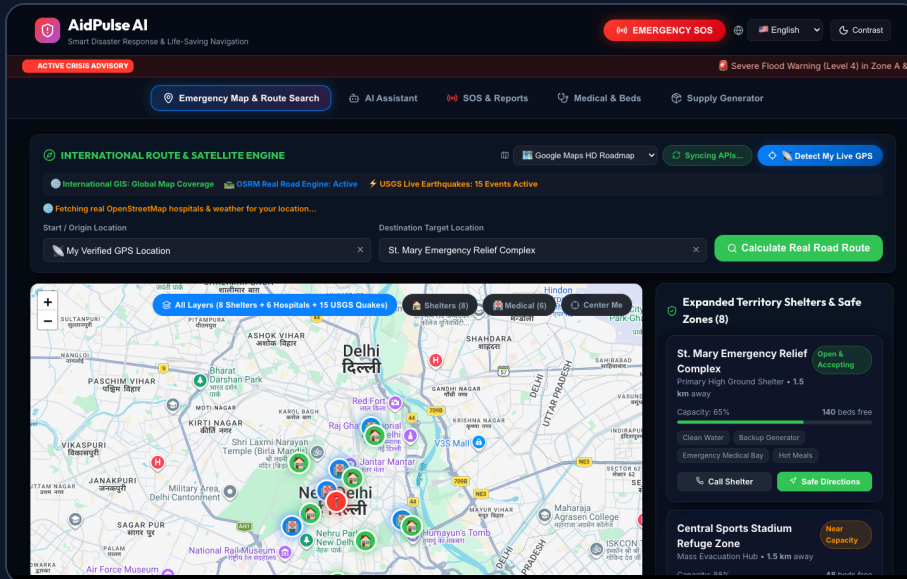


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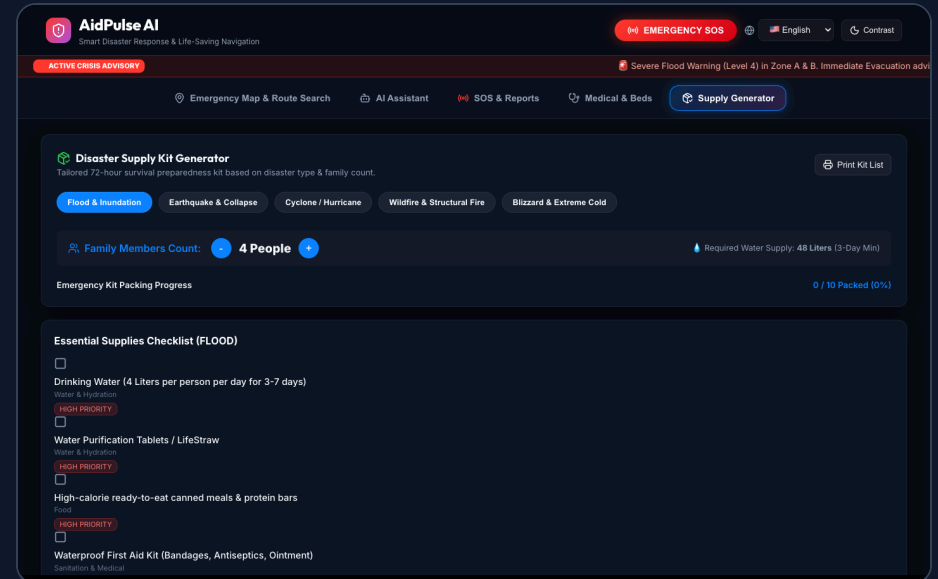
- **7-Day Satellite Risk Predictor:** Computes location risk scores (LOW, MODERATE, HIGH, CRITICAL).
- **Gemini 1.5 Flash Triage:** Answers disaster queries with actionable emergency guidance.
- **Voice & Hands-Free Audio:** Speech recognition input and text-to-speech audio readout.
- **Embedded Action Cards:** Triggers instant SOS dispatch and shelter directions inside chat.

FEATURE SHOWCASE 3

Hospital ER Tracker & Emergency Survival Calculator



Hospital ER & ICU Bed Tracker



Family Survival Supply Calculator

Production Architecture & Global Scalability

Technical Architecture

- **Frontend:** React 18, Vite 8, Leaflet GIS, Lucide Icons.
- **AI Engine:** Google Gemini 1.5 Flash REST API (`gemini-1.5-flash`).
- **GIS & Routing:** OSRM Engine, OpenStreetMap Overpass & Nominatim.
- **Telemetry:** USGS Earthquake Feed & Open-Meteo Weather API.

Scalability & Future Vision

- **Lightning Speed:** Vite production bundle builds in ~130ms.
- **Offline Resilience:** Client-side rule engine fallbacks if APIs disconnect.
- **IoT Drone Rescue:** Integrating payload dispatch telemetry for disaster zones.
- **Satellite Mesh Sync:** Enabling P2P offline mesh emergency sync.